

Uniform Derivative Envelopes for Projection-Free $\det_2$ Tails

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July 2026

Abstract

The modulus-complete complement of RH-282 has an all-order geometric trace envelope. We show that the same estimate controls every fixed derivative of the omitted logarithmic determinant tail. If $|\tau_{\sigma,n}| \leq M_\sigma q^{n-2}$, $M_\sigma \leq C\sigma^{-\alpha}$, and $qR < 1$, then the s th derivative of $\sum_{n \geq m} \tau_{\sigma,n} z^n / n$ on $|z| \leq R$ is bounded by a constant times $M_\sigma m^{s-1} (qR)^m$. A logarithmic clock strictly above the sharp RH-283 threshold therefore gives convergence in every fixed C^s norm. In the Hardy-disk instance the gain is a positive power of noise. This theorem controls only the complementary spectral factor; the finite head-to- monodromy bridge remains open.

1 Derivative estimate

Assume

$$|\tau_{\sigma,n}| \leq M_\sigma q^{n-2}, \quad n \geq 2,$$

and define

$$E_{\sigma,m}(z) = \sum_{n \geq m} \frac{\tau_{\sigma,n}}{n} z^n.$$

Theorem 1.1. *Let $x = qR < 1$ and fix an integer $s \geq 0$. There is a finite constant*

$$B_{s,x} = \sum_{k \geq 0} (1+k)^{\max(s-1,0)} x^k$$

such that, for $m \geq \max(2, s)$,

$$\sup_{|z| \leq R} |E_{\sigma,m}^{(s)}(z)| \leq M_\sigma q^{-2} R^{-s} B_{s,x} m^{\max(s-1,0)} x^m. \quad (1)$$

Proof. For $s = 0$, use $1/n \leq 1/m$. For $s \geq 1$, differentiating z^n/n gives a coefficient no larger than $n^{s-1} R^{n-s}$. Write $n = m + k$ and use $(m+k)^{s-1} \leq m^{s-1} (1+k)^{s-1}$. The remaining series is exactly $B_{s,x}$. $\square$

Corollary 1.2 (smooth tail convergence). *If $M_\sigma \leq C\sigma^{-\alpha}$ and $m_\sigma = \lceil a \log(1/\sigma) \rceil$ with $a \log(1/(qR)) > \alpha$, then*

$$E_{\sigma,m_\sigma} \longrightarrow 0 \quad \text{in } C^s(|z| \leq R)$$

for every fixed s . The canonical-product tail $\exp(-E_{\sigma,m_\sigma})$ converges to 1 in the same topology.

Proof. The right side of (1) is a fixed power of $\log(1/\sigma)$ times $\sigma^{a \log(1/(qR)) - \alpha}$. It tends to zero. The exponential claim follows from the chain rule on a bounded neighborhood of zero. $\square$

2 Certified instance

For $\alpha = 1$, $q = 1/2$, $R = 7/5$, and $a = 4$, the exponent is

$$4 \log(10/7) - 1 = 0.426699775 \dots > 0.$$

Thus the high-order noisy spectral factor from RH-282 tends to one together with every fixed derivative on the full target disk.

Remark 2.1. Derivative control of the tail cannot identify the finite head. In particular it does not show that the modulus-selected noisy roots lie on the finite monodromy shell. Gates A–E remain false/open, and no arithmetic or Hilbert–Polya conclusion is drawn.